

OLIVER BOORSTEIN

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Education

University of Oregon

Sep 2023 – Jun 2027

B.S. in Computer Science and Data Science

Eugene, Oregon

Minors: Mathematics, Music Technology | GPA: **4.22 / 4.30**

Relevant Coursework: Parallel Computing, Databases, Deep Learning, Data Structures & Algorithms, Linear Algebra

Experience

QuackHacks

May 2026 – Present

Software Engineer and Event Director

Eugene, Oregon

- Led the University of Oregon's largest hackathon with over **200** hackers and a budget of **\$50,000**.
- Built production **Next.js** and **PostgreSQL** app sustaining over **10,000,000** realtime messages across **20+** tables.
- Deployed an **AWS S3** and **CloudFront** image pipeline to serve images to the QuackHacks web platform.

Oregon Networking Research Group

May 2025 – Present

Undergraduate Researcher

Eugene, Oregon

- Architected **ATLANTIS**: a **SvelteKit**, **TypeScript**, data processing and exploration platform for visualizing large-scale network telemetry, deployed on **Cloudflare Workers** with **D1** and **Drizzle ORM**.
- Built a parallelized **Rust** pipeline, turning **100+ TB** NetFlow datasets to **SQLite** tables for efficient analytic queries.
- Ported address-analysis routines from **Haskell** to **C++**, reducing processing time by **80%**.
- Developed **PyTorch GRU** pipeline to forecast networking, reducing **MAE** by **10%** over traditional methods.

Album Listening Club

Jun 2024 – Present

Co-President

Eugene, Oregon

- Led weekly meetings and a **\$2,000** budget for one of Oregon's largest student clubs, now **50+** weekly attendees.
- Directed local music events generating over **\$5,000** in revenue and over **1,000** attendees across events.

Projects & Open Source

T3 Code | *React, TypeScript, Electron, Effect, Vitest*

Aug 2026 – Present

- Contributed **8** merged PRs to **T3 Code**, Theo Browne's coding-agent app with **21k+** GitHub stars and **50k+** users.
- Built SSH host autocomplete and fixed markdown rendering, theme contrast, and PR tab state in the **Electron** app.

tagium | *React, TypeScript, Cloudflare Workers, Fly.io*

Aug 2025 – Present

- Deployed **tagium.app**, an efficient browser-local music tag editor with **1,100+** tracks tagged by **100+** users.
- Designed a format-detecting lossless metadata editing engine for MP3, FLAC, and M4A all behind a single interface.
- Ran a horizontally scaled download queue with capped concurrency and retries, resulting in error rates below **0.1%**

Engage Form Assistant | *SvelteKit, Convex, Clerk, TypeScript*

Apr 2026 – Present

- Built a **SvelteKit** and **Convex** web app and browser extension automating **15-step** university purchase requests.

L-Store Database | *Python, pytest*

Sep 2025 – Dec 2025

- Directed a team of 4 to build a lineage-based storage engine in **Python** (**80%+** test coverage via **pytest**).
- Designed a concurrent transaction system using strict 2-phase locking, preventing deadlocks and ensuring thread-safety.

Awards

Vice President for Research and Innovation Fellowship

Jun 2026 – Sep 2026

\$5,000 research fellowship

University of Oregon

Bowerman Scholarship

Jun 2026

\$9,000 nomination-based award

University of Oregon

Geoffrey Eric Wright Outstanding Junior Award

May 2026

\$3,000 sole recipient

University of Oregon Department of Computer Science

Technical Skills

Languages: TypeScript, JavaScript, Python, SQL, Rust, C/C++, Bash

Frontend: React, Svelte/SvelteKit, Next.js, Astro, Tailwind CSS, Vite, HTML, CSS

Backend & Data: Convex, PostgreSQL, SQLite, Supabase, Drizzle ORM

Infrastructure: Cloudflare (Workers, D1, R2), AWS (S3, CloudFront), Fly.io, Git, GitHub Actions, Bun, Node.js, Linux

Testing: Playwright, Vitest, pytest